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ASSIGNMENTS

# Problem set 1

AI504 — Knowledge Representation

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## Problem 1

An *interval* is an ordered pair of real numbers  $(a, b)$  such that  $a < b$ . We say that the interval *interleaves* the interval  $(c, d)$  in case  $a < c < b < d$ . Is the “interleaves” relation transitive? If not, what is its transitive closure?

## Solution

The relation definition is that

$$\text{if } (x, y) \in R \text{ then } xRy.$$

The interleaves relation is defined as:

$$(a, b)R(c, d) = a < c < b < d.$$

For transitivity, it is required that

$$(a, b)R(c, d) \text{ and } (c, d)R(e, f) \Rightarrow (a, b)R(e, f).$$

By the interleaves definition, we know that

$$(a, b)R(c, d) = a < c < b < d$$

$$(c, d)R(e, f) = c < e < d < f$$

we can call these our premises.

Now we wish to prove that

$$(a, b)R(e, f) = a < e < b < f.$$

For this, there are the following requirements

$$a < c < e < b < d < f$$

then by our premises we can prove the following

$$a < c \text{ and } c < e \Rightarrow a < e$$

and

$$b < d \text{ and } d < f \Rightarrow b < f.$$

Since we only know that

$$c < e < d \text{ and } c < b < d$$

we cannot say anything about whether  $e < b$  is true.

Therefore, the relation is **not transitive**.

Let's do a **counterexample**.

Let

$$A = [1, 3] \quad B = [2, 6] \quad C = [5, 8].$$

If we apply

$$ARB \text{ and } BRC \Rightarrow ARC.$$

This would imply that  $[1, 3]$  interleaves  $[5, 8]$ .

This is false because it would imply that

$$1 < \mathbf{5} < \mathbf{3} < 8.$$

For this to work we need the relation closure  $R^+$ .

Let's argue that for the closure, we need

$$(a, b)R(c, d) \text{ and } (c, d)R(e, f) \Rightarrow (a, b)R(e, f)$$

where

$$(a, b)R(e, f) = a < e < b < f$$

Since by definition we already know  $a < e$  and  $b < f$ .

The closure needs to be

$$R^+ = \{(e, b)\}.$$

## Problem 2

Let  $x$  and  $Y$  be sets, let  $R$  and  $S$  be binary relations of type  $X \times Y$ , and let  $P$  and  $Q$  be subsets of  $X$ . Recall that the *forward image*  $R(P)$  is the subset of  $Y$  obtained by following all  $R$ -arrows forwards out of  $P$ . (And similarly for all the other forward images in this problem.) For each of the following identities, either prove correct or provide a counterexample. *All proofs are short and all counterexamples are tiny*

(a)  $R(P \cup Q) = R(P) \cup R(Q)$

(b)  $R(P \cap Q) = R(P) \cap R(Q)$

(c)  $(R \cup S)(P) = R(P) \cup S(P)$

(d)  $(R \cap S)(P) = R(P) \cap S(P)$

## Solution

A relation is a relationship between sets of values such that for a set  $X$

$$R \subseteq \{(x, y) \mid x, y \in X\}$$

(a)

We wish to show that

$$R(P \cup Q) = R(P) \cup R(Q)$$

We know that

$$\forall y \in R(P \cup Q), \exists x \in P \cup Q \mid (x, y) \in R$$

This means that for every  $y$  in the relation on the union of the subsets  $P$  and  $Q$ , there must be some  $x$  where the relation  $R$  holds.

Then we can show that the  $y$  can either come from  $R(P)$  or  $R(Q)$ , so

$$\forall y \in R(P) \cup R(Q) : y \in R(P) \text{ or } y \in R(Q).$$

This is the same as  $R(P \cup Q) = R(P) \cup R(Q)$ .

□

(b)

We wish to show that

$$R(P \cap Q) = R(P) \cap R(Q).$$

The same applies for

$$\forall y \in R(P \cap Q), \exists x \in P \cap Q \mid (x, y) \in R$$

again the  $y$  must come from an  $x$  mapped  $R$ .

But when we do

$$\forall y \in R(P) \cap R(Q) : y \in R(P) \text{ and } y \in R(Q).$$

We realise that for  $y$  to exist,  $P$  and  $Q$  must share an element that both maps to the same  $y$ , if they don't then **this does not hold**.

Let's do a **counterexample** to properly disprove this.

Let's check if

$$R(P \cap Q) = R(P) \cap R(Q)$$

when

$$X = \{1, 2\} \text{ and } Y = \{1, 2\} \text{ and } P = \{1\} \subseteq X \text{ and } Q = \{2\} \subseteq X.$$

Now let

$$R = \{(1, 2), (2, 2)\}$$

then

$$R(P \cap Q) = \{(1, 2), (2, 2)\}(\{1\} \cap \{2\}) = \emptyset$$

and

$$R(P) \cap R(Q) = \{(1, 2), (2, 2)\}(\{1\}) \cap \{(1, 2), (2, 2)\}(\{2\}) = \{2\} \cap \{2\} = \{2\}.$$

Since  $\emptyset \neq \{2\}$  this **counters** the statement.

□

(c)

We wish to show that

$$(R \cup S)(P) = R(P) \cup S(P)$$

we can say that any  $y$  must come from an  $x$  in  $P$  applied by  $R$  or  $S$

$$\forall y \in (R \cup S)(P), \exists x \in P \mid (x, y) \in R \text{ or } S.$$

Then we can describe  $y \in R(P) \cup S(P)$

$$\forall y \in R(P) \cup S(P), \exists x \in P \mid (x, y) \in R \text{ or } S.$$

Since these both describe the same relation from  $x \rightarrow y$  they are **equal**.

□

(d)

We wish to show that

$$(R \cap S)(P) = R(P) \cap S(P)$$

any  $y$  must have an  $x$  from  $P$  which is applied by  $R \cap S$

$$\forall y \in (R \cap S)(P), \exists x \in P \mid (x, y) \in R \cap S.$$

Then showing that  $x$  can only be

$$\forall y \in R(P) \cap S(P), \exists x \in P \mid (x, y) \in R \text{ and } S.$$

Much like in (b) we realise that for  $y$  to exist,  $R$  and  $S$  must share an element that both maps the same  $x$  to the same  $y$ , if they don't then **this does not hold**.

Let's do a **counterexample** to disprove this.

Let's check if

$$(R \cap S)(P) = R(P) \cap S(P)$$

$$X = \{1, 2, 3\} \text{ and } Y = \{1, 2, 3\} \text{ and } P = \{1, 2\} \subseteq X.$$

Now let

$$R = \{(1, 2), (1, 3), (3, 1)\} \quad S = \{(1, 3), (2, 1), (2, 2), (3, 3)\}$$

then

$$R \cap S = \{(1, 3)\}.$$

Now we can check

$$(R \cap S)(P) = \{(1, 3)\}(\{1, 2\}) = \{3\}$$

and

$$\begin{aligned} R(P) \cap S(P) &= \{(1, 2), (1, 3), (3, 1)\}(\{1, 2\}) \cap \{(1, 3), (2, 1), (2, 2), (3, 3)\}(\{1, 2\}) \\ &= \{2, 3\} \cap \{1, 2, 3\} = \{2, 3\}. \end{aligned}$$

Since  $\{3\} \neq \{2, 3\}$  this **counters** the statement.

□

### Problem 3

A *full binary tree* is a nonempty binary tree where each node is a leaf or has exactly two children. Define a function  $j$  from full binary trees to rational numbers by recursion as follows:

$$j(T) := \begin{cases} 1 & \text{if } T \text{ is a leaf} \\ 1 + \frac{j(T_0) + j(T_1)}{4} & \text{if otherwise, where } T_0 \text{ and } T_1 \text{ are the two subtrees} \end{cases}$$

Calculate the value of  $j$  on lots of small examples. Formulate a conjecture about how big it can get, and prove it.

### Solution

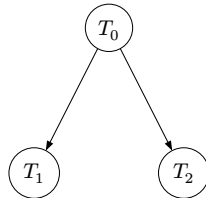
Let's start off by doing a few examples

#### Example 1



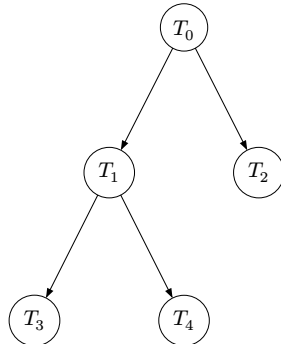
Here  $j(T) = 1$

#### Example 2



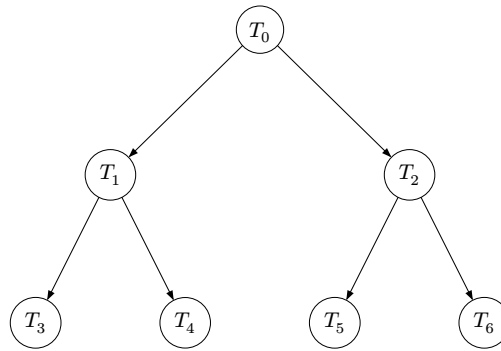
Here  $j(T) = 1 + \frac{1+1}{4} = 1 + \frac{1}{2} = 1.5$

#### Example 3

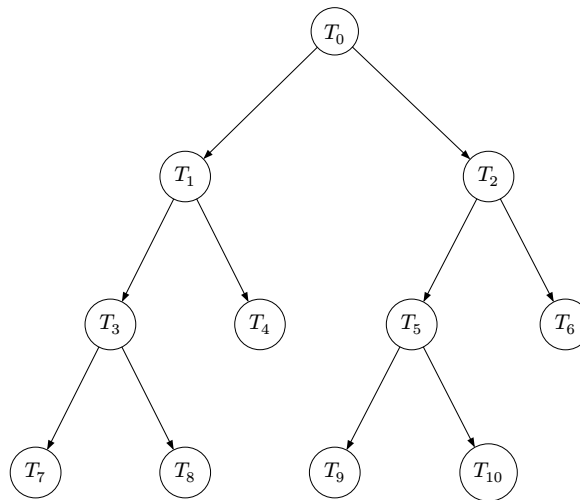


Here  $j(T) = 1 + \frac{(1 + \frac{1+1}{4}) + 1}{4} = 1 + \frac{2 + \frac{1}{2}}{4} = 1.625$



**Example 4**

Here  $j(T) = 1 + \frac{(1+\frac{1+1}{4})+(1+\frac{1+1}{4})}{4} = 1 + \frac{3}{4} = 1.75$

**Example 5**

Here  $j(T) = 1 + \frac{\left(1 + \frac{(1+\frac{1+1}{4})+1}{4}\right) + \left(1 + \frac{(1+\frac{1+1}{4})+1}{4}\right)}{4} = 1 + \frac{13}{16} = 1.8125$

## Conjecture

From the examples above, it seems that  $j(T_{\text{size } k}) \rightarrow 2$  as  $k \rightarrow \infty$ .

Therefore, for any finite tree  $T$ , it seems that  $j$  grows infinitely close to 2. Therefore:

$$j(T) < 2$$

Let's try and prove that

### Proof by structural induction

We want to prove that  $j(T) < 2$

#### Base case

if  $T_0$  is a single leaf (a node with no children) then  $j(T) = 1$

$$j(T_0) < 2.$$

#### Inductive hypothesis

Assume that

$$j(T_{\text{subtree } 1}) < 2 \text{ and } j(T_{\text{subtree } 2}) < 2$$

#### Inductive step

For any tree with two subtrees (full tree)

$$j(T) = 1 + \frac{j(T_{\text{subtree } 1}) + j(T_{\text{subtree } 2})}{4}.$$

Since, by **IH**

$$j(T_{\text{subtree } 1}) + j(T_{\text{subtree } 2}) < 4.$$

A new tree, composed of two subtrees, becomes

$$j(T) = 1 + \frac{j(T_{\text{subtree } 1}) + j(T_{\text{subtree } 2})}{4}.$$

Since  $\frac{j(T_{\text{subtree } 1}) + j(T_{\text{subtree } 2})}{4} < 1$

$$j(T) = 1 + \frac{j(T_{\text{subtree } 1}) + j(T_{\text{subtree } 2})}{4} < 2$$

## Problem 4

Consider a directed graph  $G$  with 12 vertices that satisfies the following two properties:

- $G$  is *strongly connected*, i.e., between any two distinct vertices  $u$  and  $v$ , there is a directed path from  $u$  to  $v$  as well as a directed path from  $v$  to  $u$ , and
- there is at most one directed edge coming out of each vertex.

Draw  $G$ , and argue why your drawing is the *only possible* solution (up to isomorphism).

## Solution

Since each vertex has **at most one** directed edge going out, and  $G$  must be strongly connected, every vertex must have **exactly one** outgoing as well as **exactly one** ingoing edge as every node must also be reachable. This creates a 12-cycle:

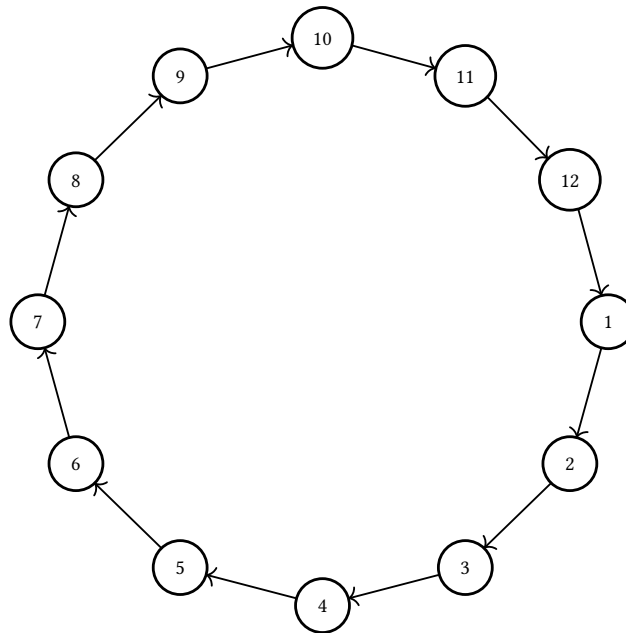


Figure 1:  $G$  : A strongly connected 12-cycle.

To say that Figure 1 is the *only possible solution* (up to isomorphism). We need to argue that Figure 1 is isomorphic.

By definition, two directed graphs  $G_1 = (V_1, E_1)$  and  $G_2 = (V_2, E_2)$  are **isomorphic** if and only if there exists a one-to-one and onto function  $f : V_1 \rightarrow V_2$  such that any two vertices  $(a, b) \in V_1$  where  $(a, b) \in E_1$  and  $(f(a), f(b)) \in V_2$  where  $(f(a), f(b)) \in E_2$ . [1]

For a function to be one-to-one and onto on two sets, it is at least required that they have the same amount of elements. For this 12-cycle, any *other solution* would, by definition, have exactly 12 vertices as well.

For this 12-cycle, because its vertices have exactly one ingoing and exactly one outgoing edge, the adjacency property of isomorphism **can** be satisfied. Any two vertices in two in a strongly connected cycle like this, will hold this property. The only way this property would **not** hold, is that a vertex in the graph has either 2 outgoing edges or 2 ingoing edges.

This can be showed with an example of 2 isomorphic graphs:

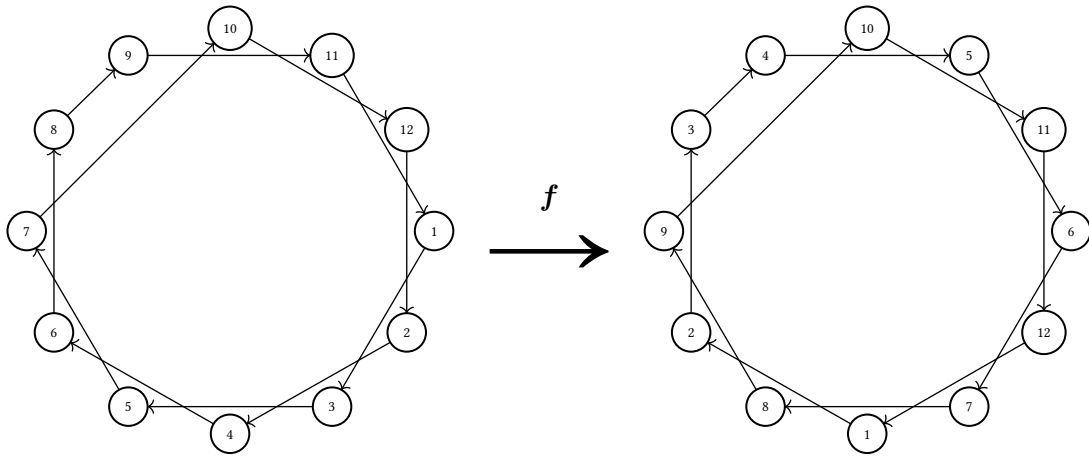


Figure 2:  $G$  : Two isomorphic directed graphs due to a function  $f$

Here, the left graph fullfills the same requirements as directed graph  $G$  on Figure 1, though it has different relations. By applying the following one-to-one and onto function  $f$  to the graph:

$$\begin{aligned} f(12) &= 11, f(2) = 12, f(4) = 1, f(6) = 2, f(8) = 3, f(9) = 4, \\ f(11) &= 5, f(1) = 6, f(3) = 7, f(5) = 8, f(7) = 9, f(10) = 10 \end{aligned}$$

yields the same relations as the original graph  $G$  on Figure 1.

Because of this, Figure 1 depicts the **only possible** solution (up to isomorphism).

## **Bibliography**

- [1] K. H. Rosen, *Discrete Mathematics and Its Applications*, 8th ed. New York, USA: McGraw-hill education, 2019.